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BEIHANG UNIVERSITY



能源与动力工程学院
School of Energy and Power Engineering



CSEET 中国工程热物理学会
Chinese Society of Engineering Thermophysics

THE 11th INTERNATIONAL CONFERENCE ON
VORTEX FLOW MECHANICS(ICVFM2025)

第十一届国际涡流力学会议

Development and Validation of a Turbulence-Chemistry Recursive Mixing Model of TPDF Simulations

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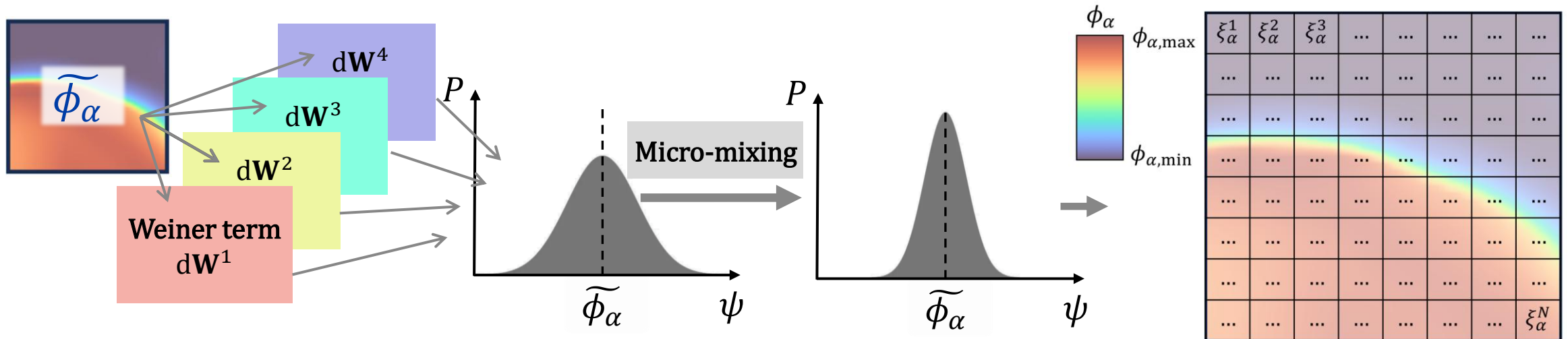
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北京航空航天大学 航空发动机数值仿真研究中心
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1. Background Introduction

□ Micro-mixing term in Eulerian stochastic field method of TPDF

$$\bar{\rho} d\xi_{\alpha}^n = -\bar{\rho}\tilde{u}_i \frac{\partial \xi_{\alpha}^n}{\partial x_i} dt + \frac{\partial}{\partial x_i} \left(\Gamma_{\alpha} \frac{\partial \xi_{\alpha}^n}{\partial x_i} \right) dt + \sqrt{2\bar{\rho}\Gamma_{\alpha}} \frac{\partial \xi_{\alpha}^n}{\partial x_i} dW_i^n + \underline{\mathcal{M}} + \bar{\rho}\dot{\omega}_{\alpha}^n dt$$

Micro-mixing term

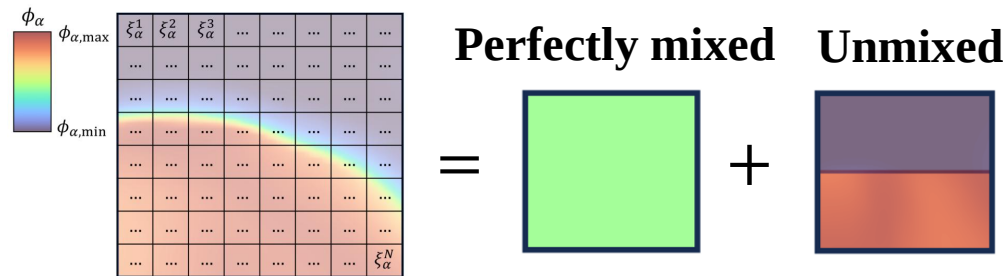


**Modify the scalar PDF formed by the Wiener term, reducing variance,
To approximate the true sampling statistics of the control volume.**

The Turbulent-Chemistry Recursive Mixing Model

Partitioning methods

- Referring to the Hybrid τ_{SGS} model and the PaSR method.
- CV is abstracted as a superposition of two reactors

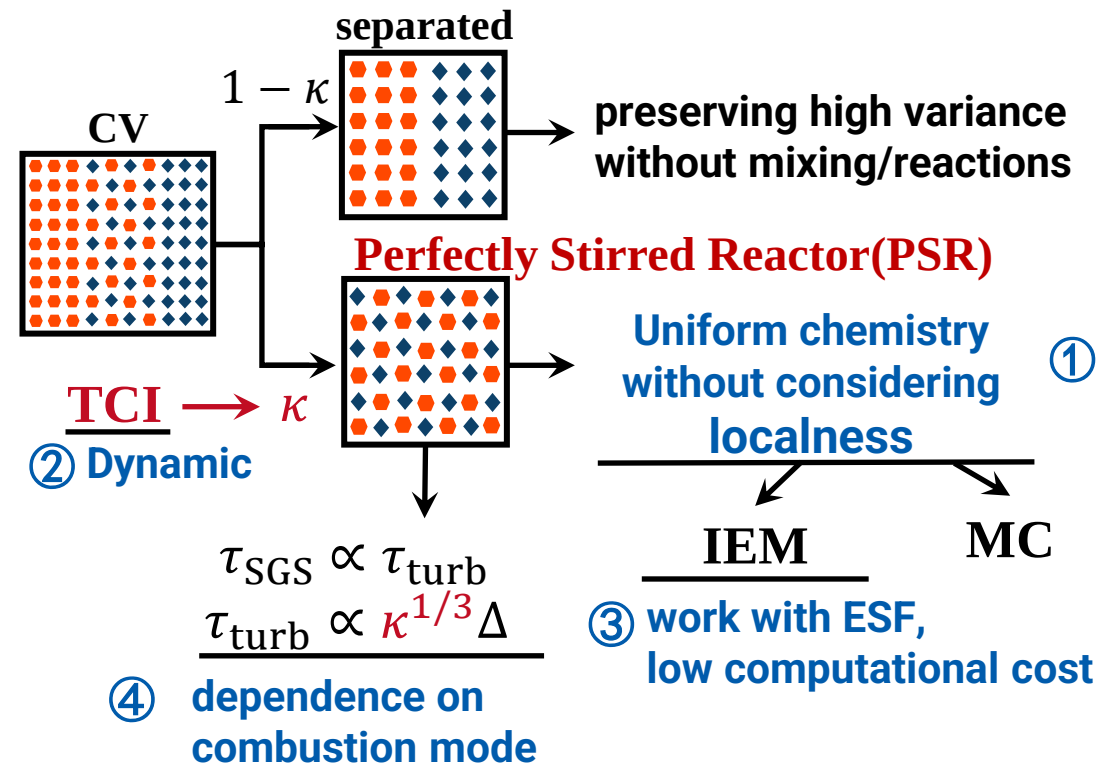


- κ_α is linked to local TCI through PaSR inspired partitioning, without timescale dependence:

$$\kappa_\alpha \sim \dot{\omega}_\alpha(\tilde{\phi}) / \dot{\omega}_\alpha(\tilde{\phi})$$

Mixing rate modeling

- Mixing is modeled in two reactors



The Multi-layer Partitioning method

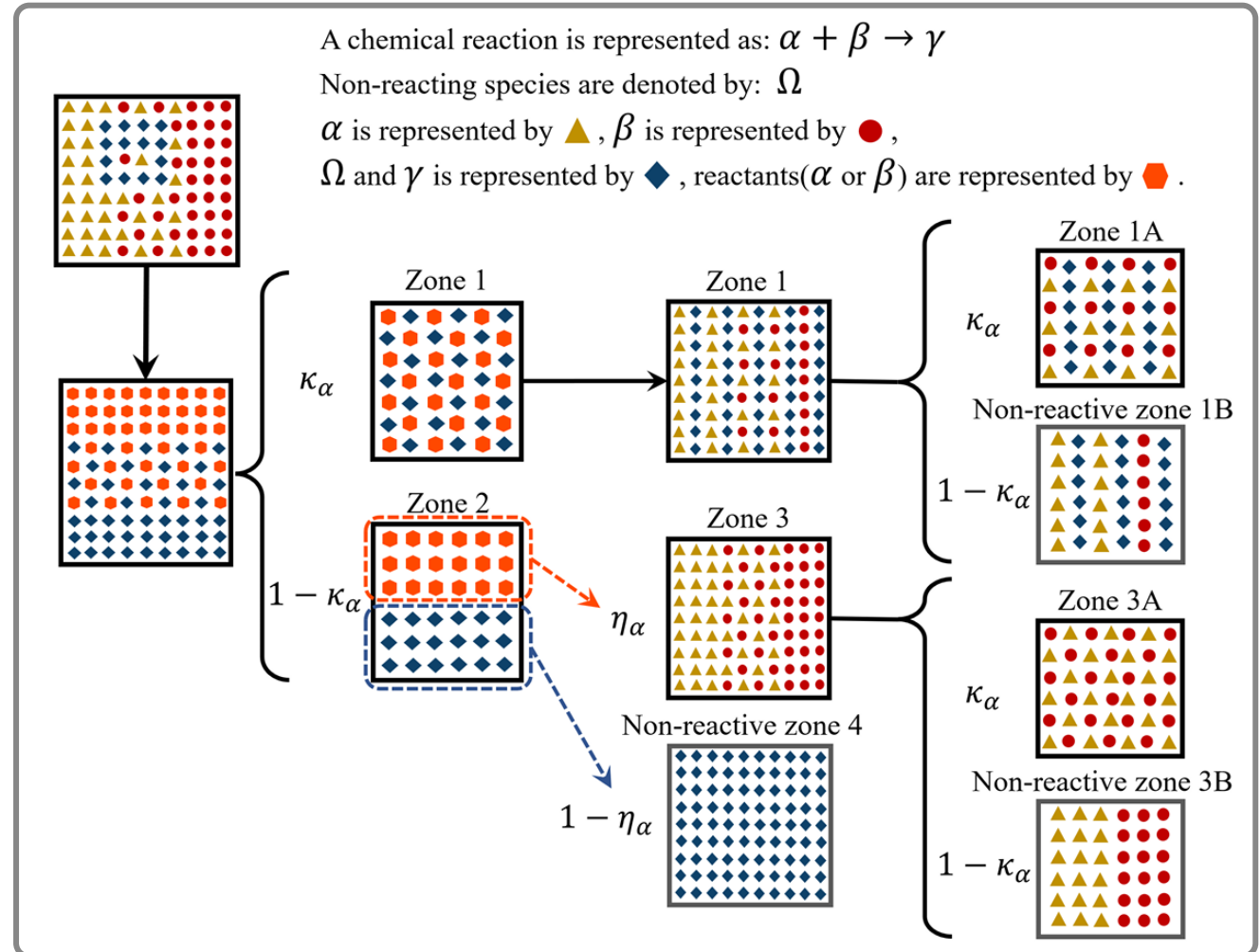
Multi-layer PaSR incorporating Dilution Effects

Correlation between filtered/laminar reaction rate:

$$\begin{aligned}
 \dot{\omega}_\alpha(\phi) &= \kappa_\alpha \dot{\omega}_1 + (1 - \kappa_\alpha) \dot{\omega}_2 \\
 &= \kappa_\alpha \dot{\omega}_1 + (1 - \kappa_\alpha) (\eta \dot{\omega}_3) \\
 &= \kappa_\alpha (\kappa_\alpha \dot{\omega}_{1A}) + (1 - \kappa_\alpha) (\eta \kappa_\alpha \dot{\omega}_{3A}) \\
 &= \left(\kappa_\alpha^2 + \frac{\kappa_\alpha (1 - \kappa_\alpha)}{\eta_\alpha} \right) \dot{\omega}_\alpha(\tilde{\phi})
 \end{aligned}$$

Correlation between mixing uniformity parameter and reaction rates :

$$\kappa_\alpha = \frac{1 \pm \sqrt{1 - 4\eta_\alpha(1 - \eta_\alpha) \frac{\dot{\omega}_\alpha(\phi)}{\dot{\omega}_\alpha(\tilde{\phi})}}}{2(1 - \eta_\alpha)}$$



The Multi-layer Partitioning method

□ Quadratic relationship between κ_α and $\widetilde{\omega}_\alpha(\phi)/\widetilde{\omega}_\alpha(\tilde{\phi})$

$$\kappa_\alpha = \frac{1 \pm \sqrt{1 - 4\eta_\alpha(1 - \eta_\alpha) \frac{\widetilde{\omega}_\alpha(\phi)}{\widetilde{\omega}_\alpha(\tilde{\phi})}}}{2(1 - \eta_\alpha)}$$

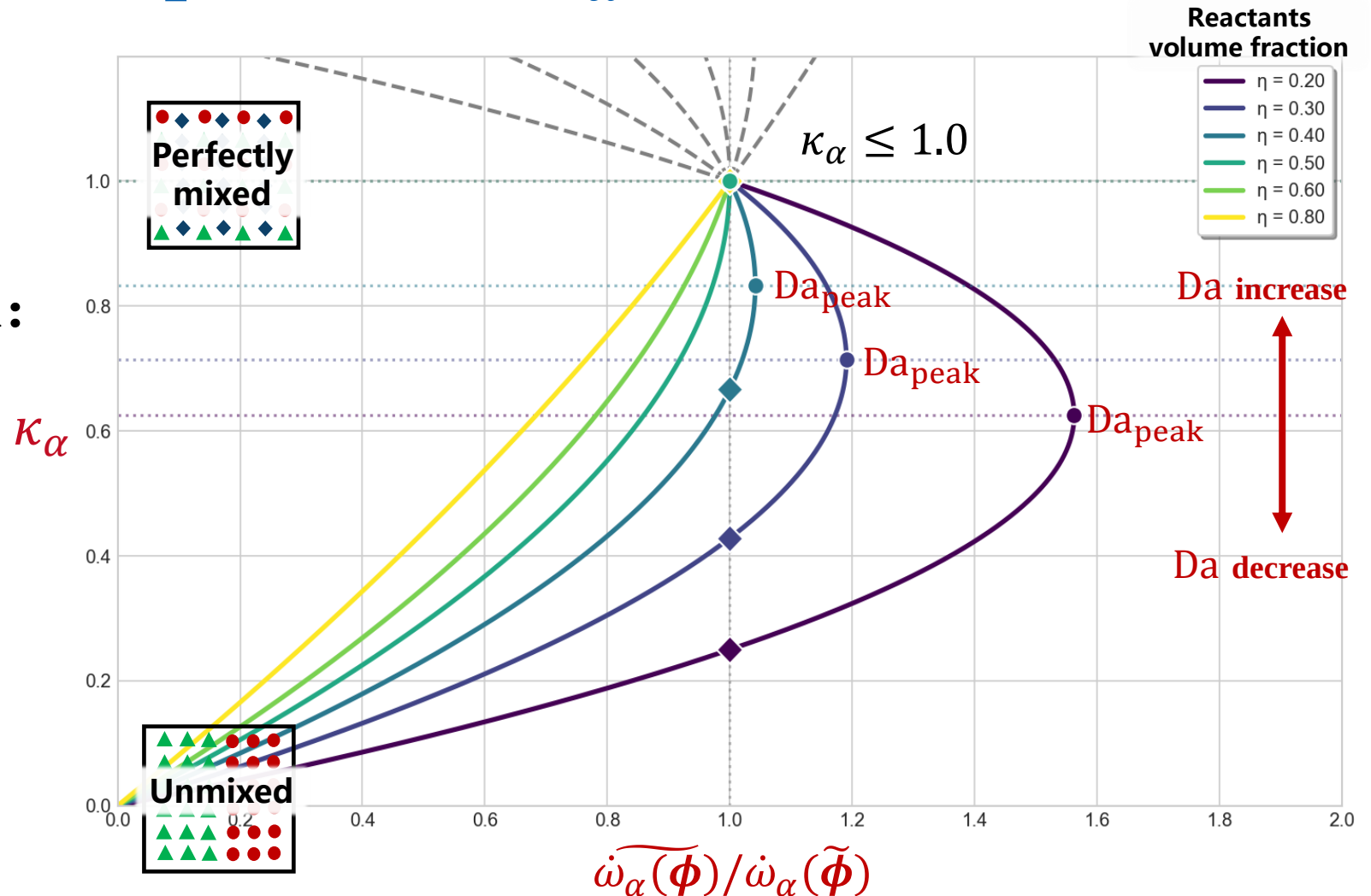
Determine $\pm$ using local Da:

Find the Da_{peak} first,

$Da > Da_{\text{peak}}, \quad \pm \rightarrow +$

$Da < Da_{\text{peak}}, \quad \pm \rightarrow -$

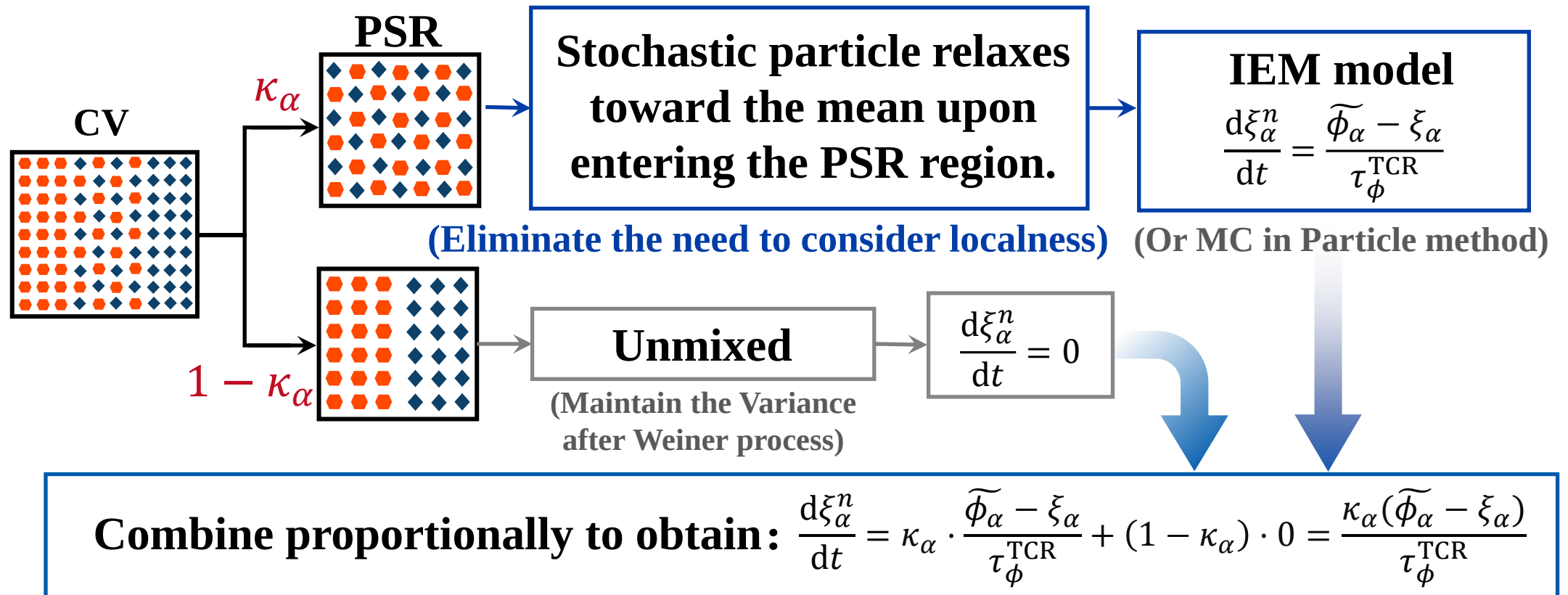
In the special case where the entire grid is filled with reactants ($\eta_\alpha = 1$), the relationship is reduced to the original PaSR formula.



The Turbulent-Chemistry Recursive Mixing Model

1. Mixing mode in the model

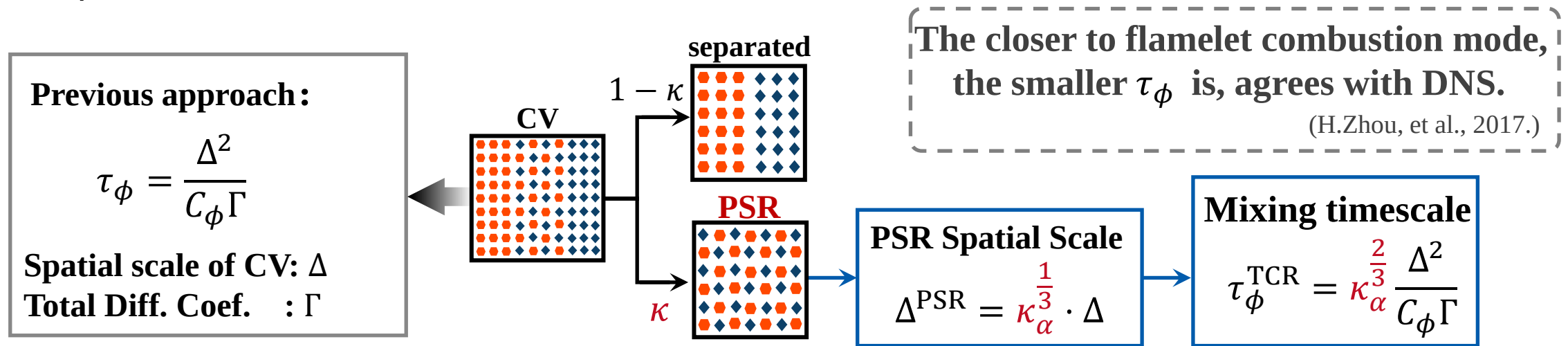
- Describe the mixing mode of components through partitioning.



The Turbulent-Chemistry Recursive Mixing Model

2. Mixing time-scale in the model

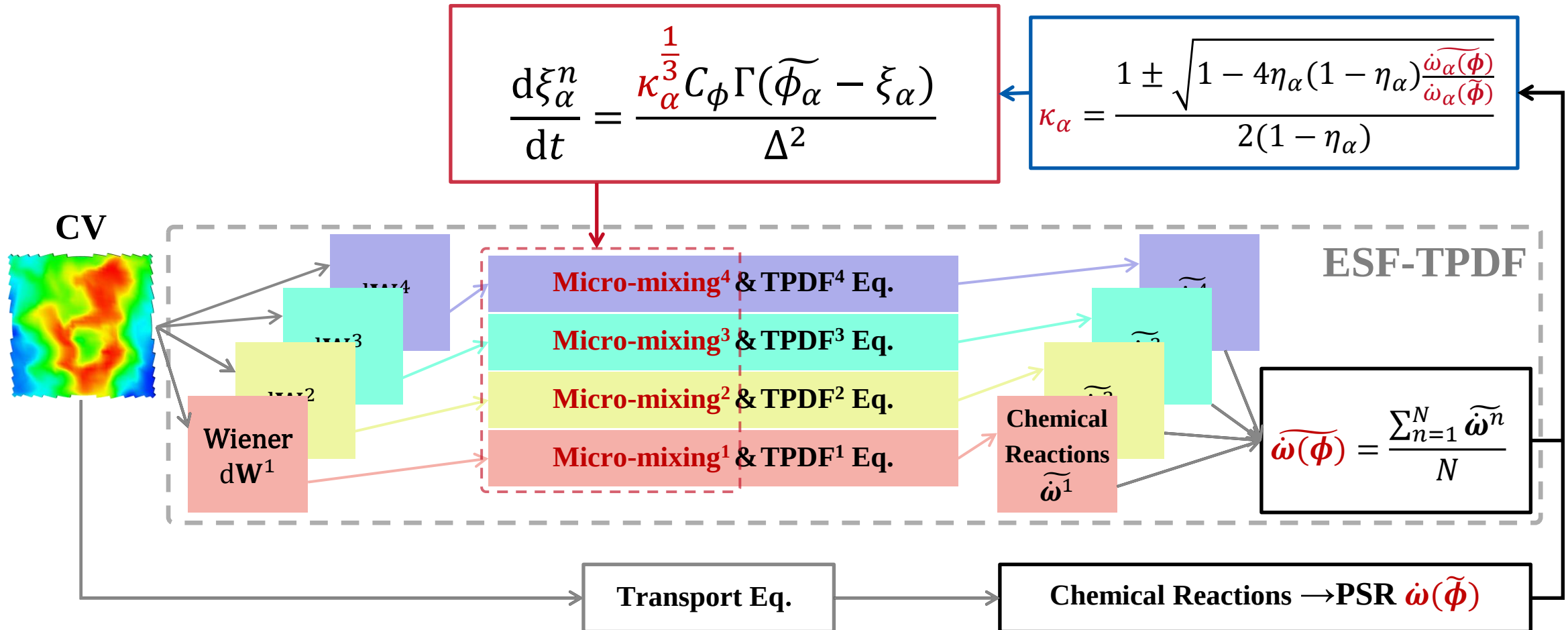
- τ_ϕ^{TCR} is calculated based on the spatial scale of the PSR



- **Final Expression of micro-mixing term in TCR model:**
$$\frac{d\xi_\alpha^n}{dt} = \frac{\kappa_\alpha (\widetilde{\phi}_\alpha - \xi_\alpha)}{\tau_\phi^{\text{TCR}}} = \frac{\kappa_\alpha^{\frac{1}{3}} C_\phi \Gamma (\widetilde{\phi}_\alpha - \xi_\alpha)}{\Delta^2}$$

The Turbulent-Chemistry Recursive Mixing Model

□ Computational Procedure for ESF using the TCR Model



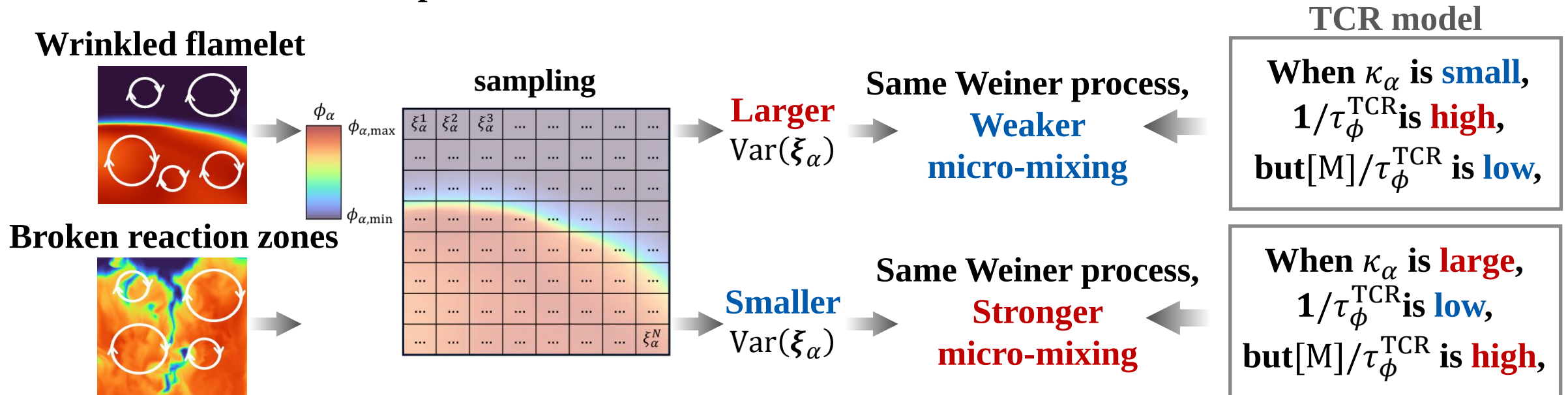
The Turbulent-Chemistry Recursive Mixing Model

□ Discussion on decay rates of $\text{Var}(\xi_\alpha)$ caused by micro-mixing

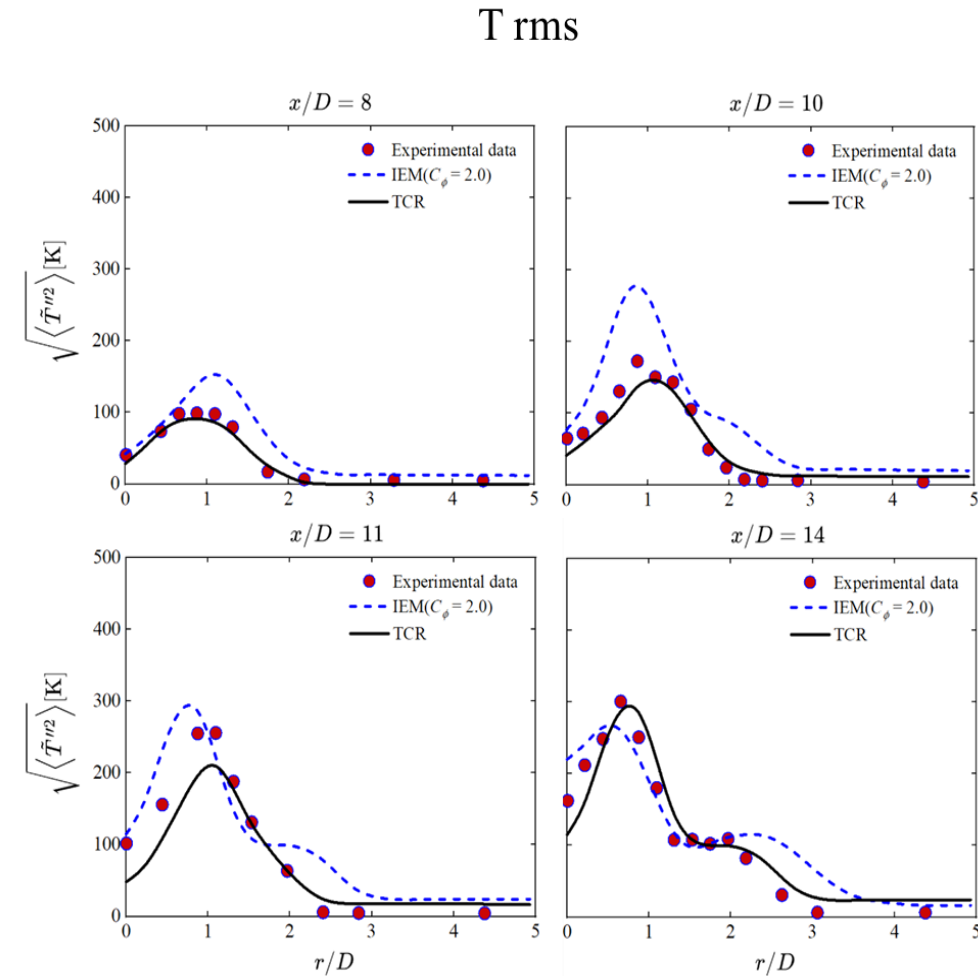
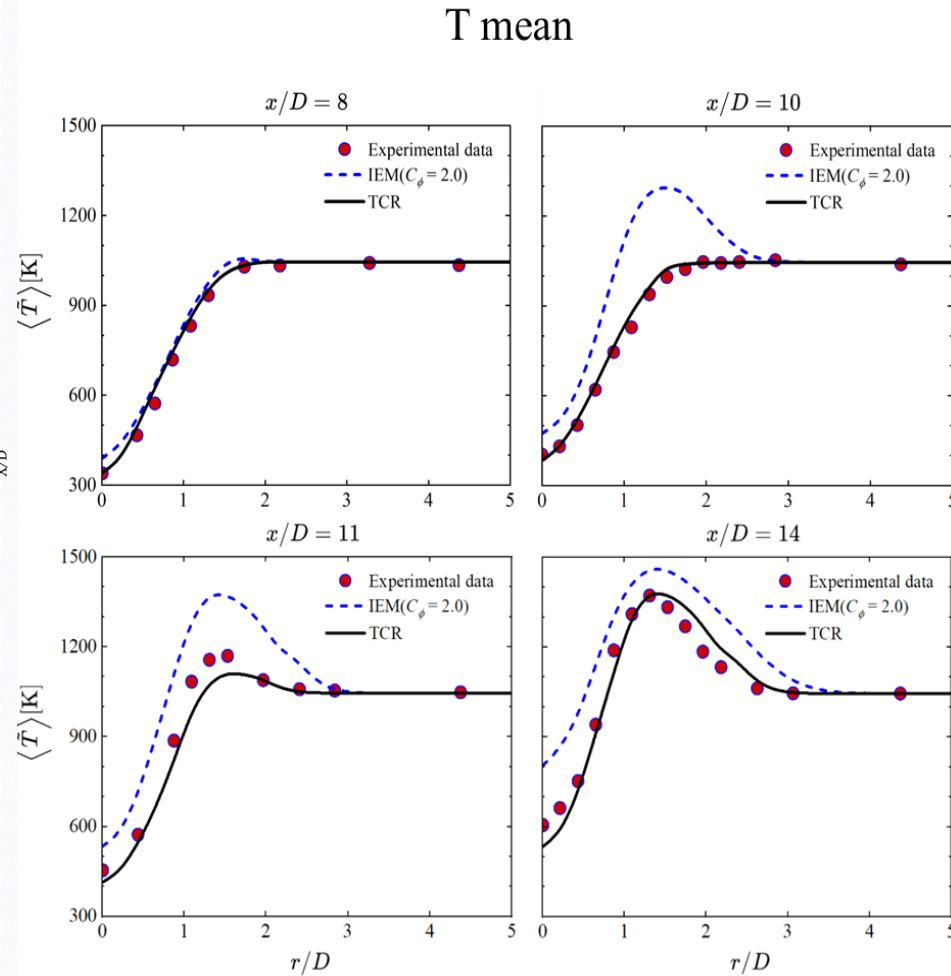
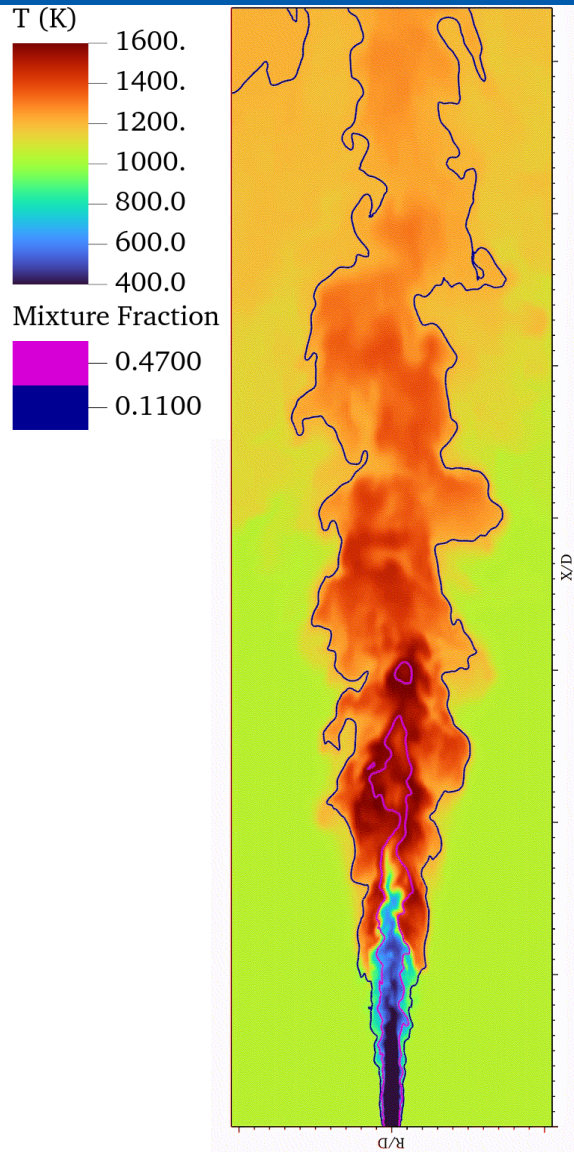
- Process causing variance changes in the ESF-TPDF method



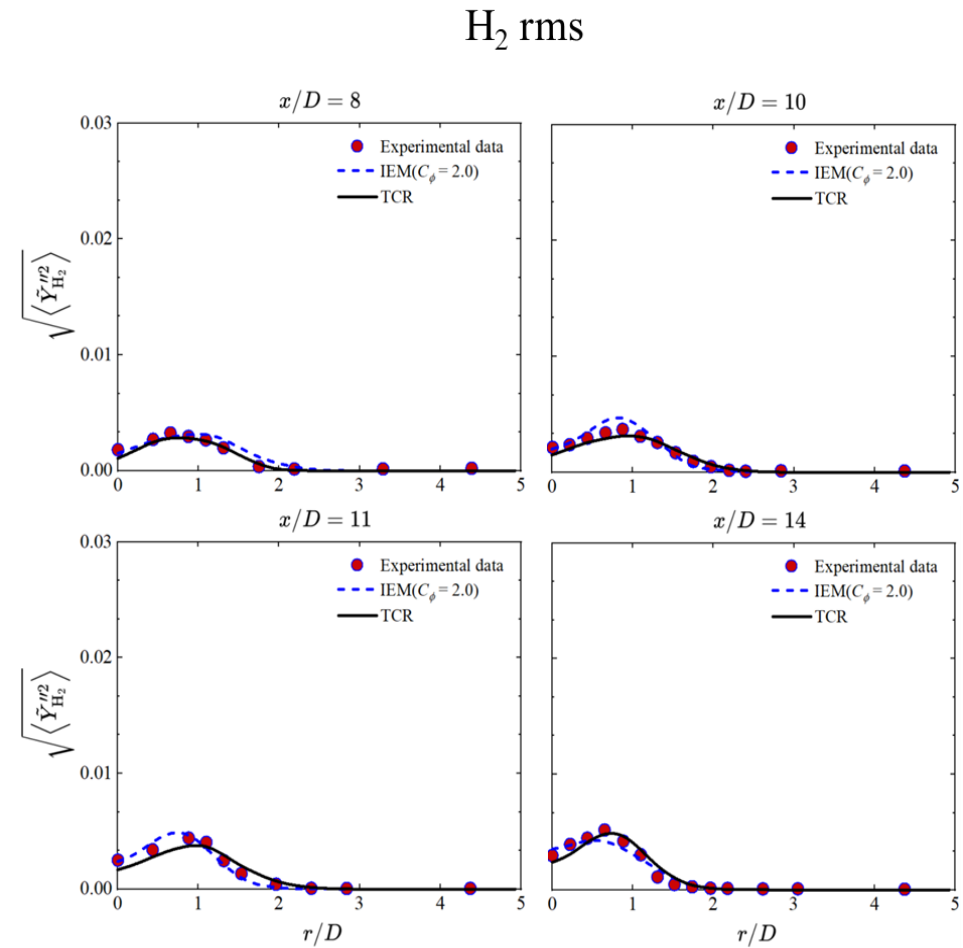
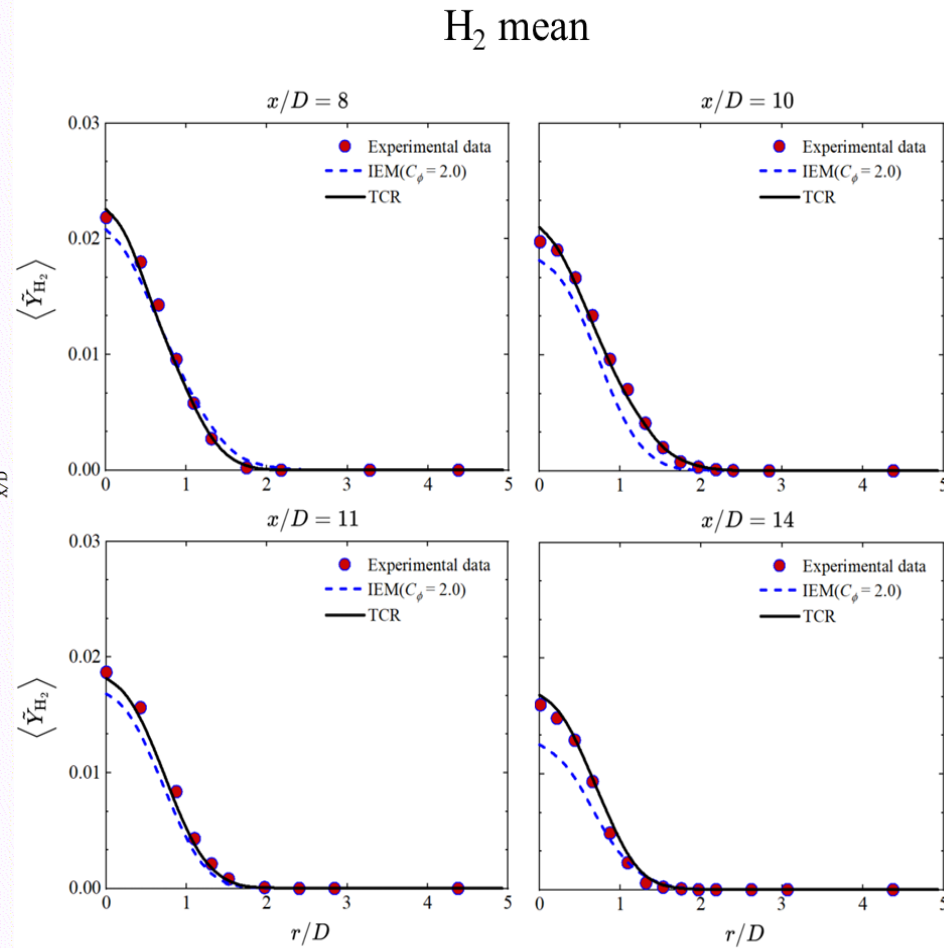
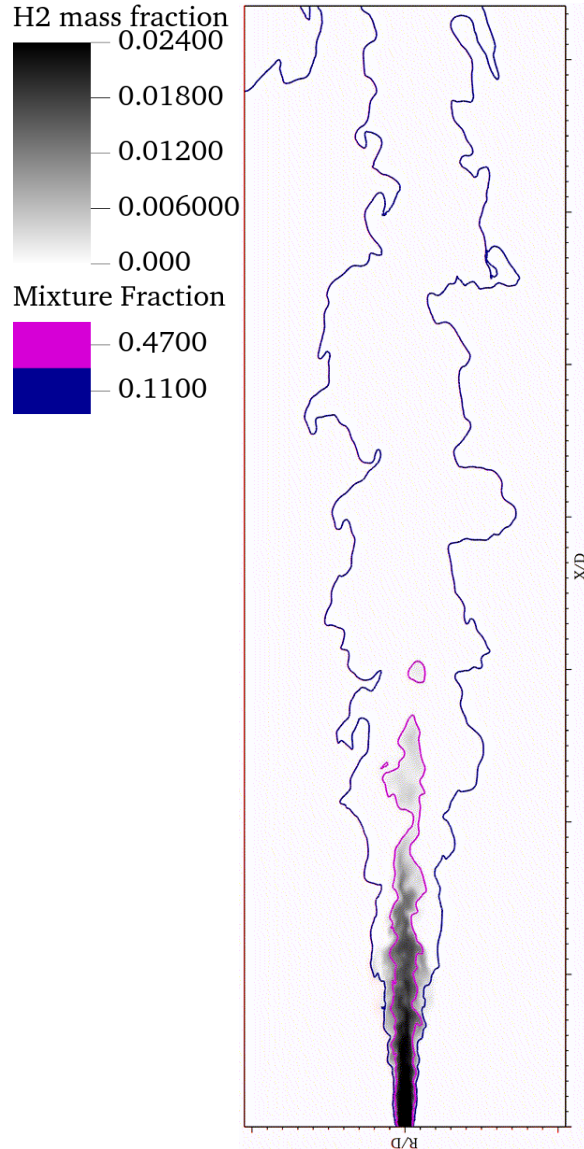
- Scalar variance sampled from combustion fields under different combustion modes



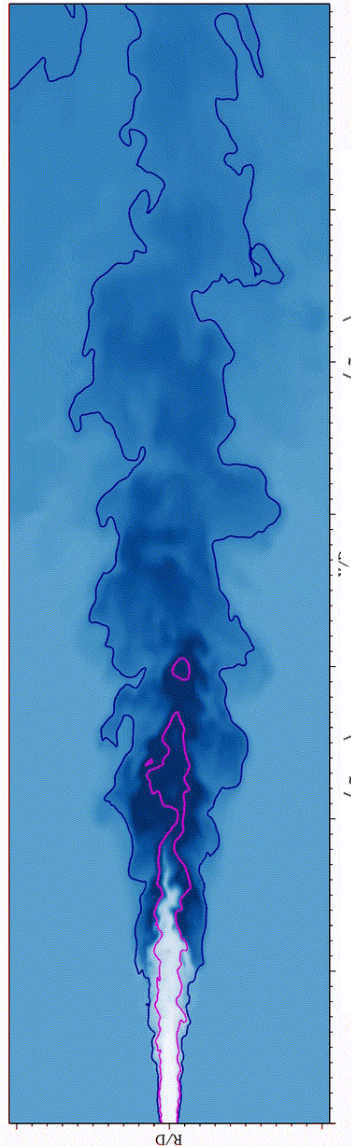
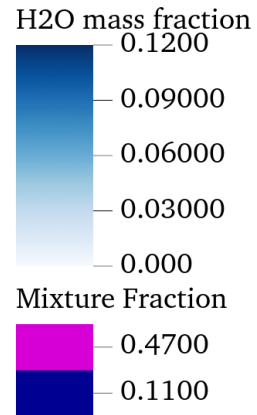
Validation in Cabra's lifted flame



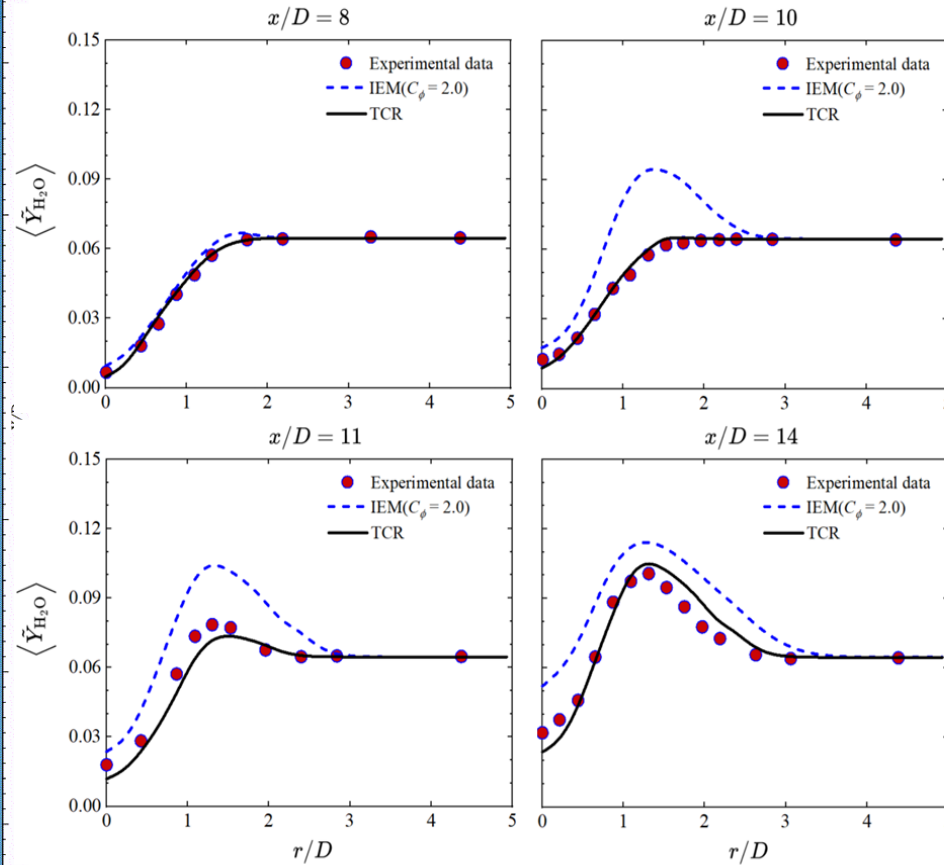
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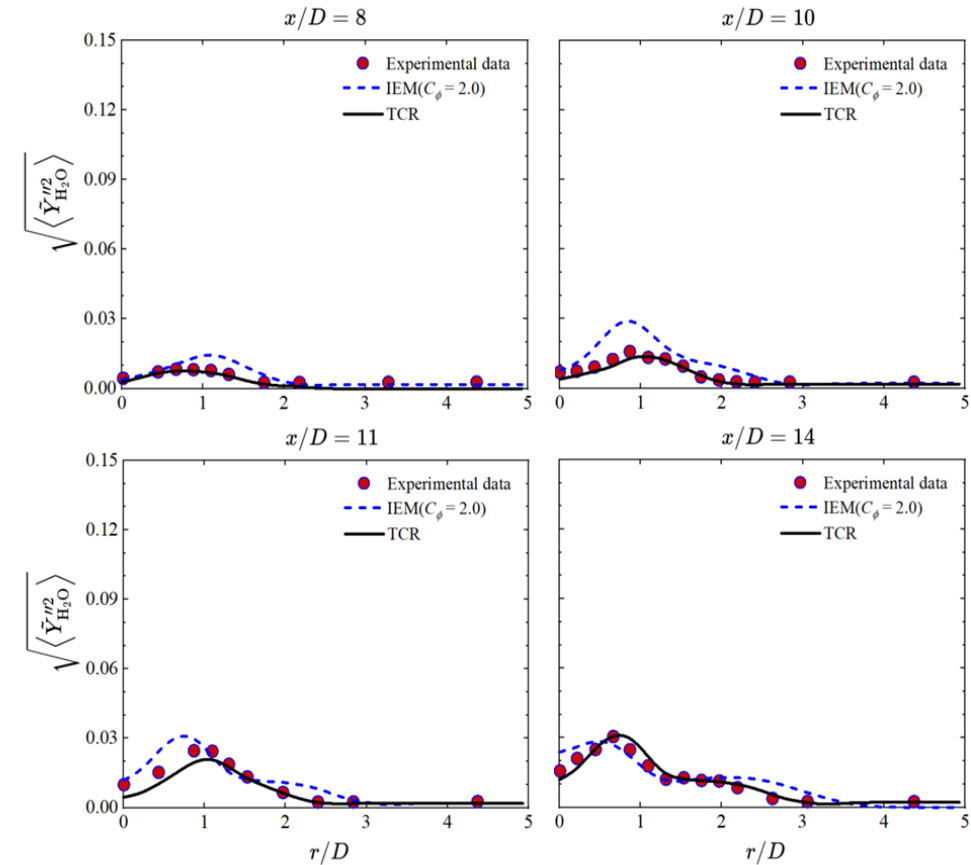
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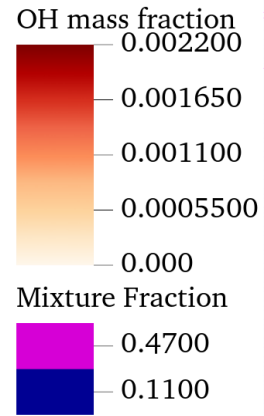
H₂O mean



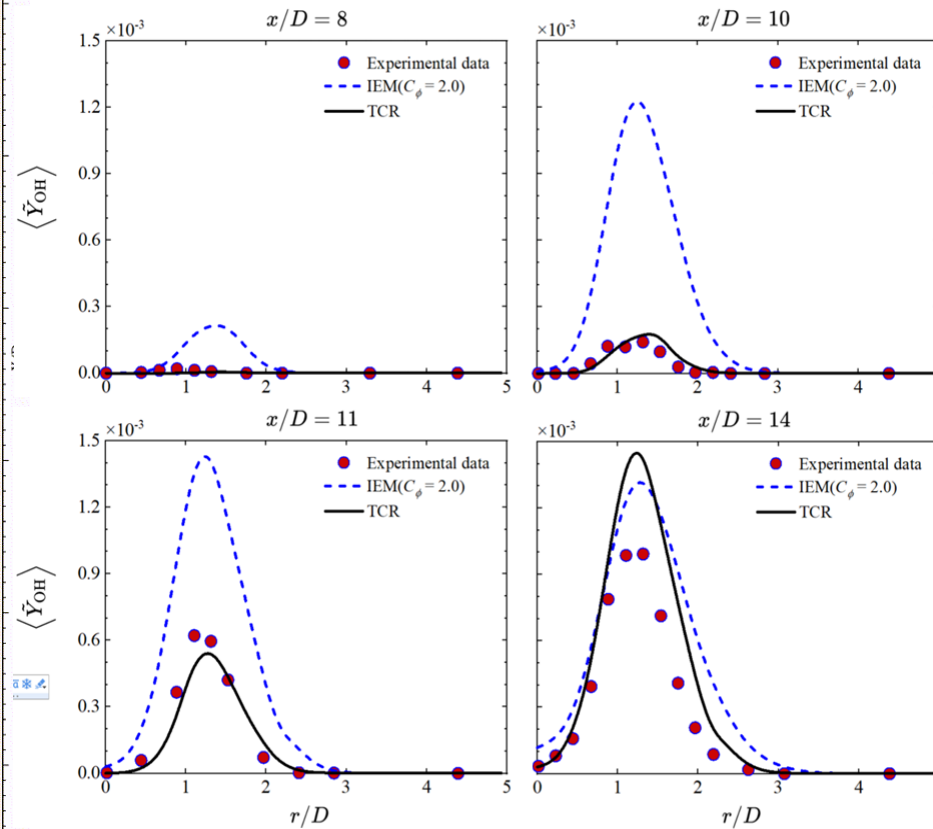
H₂O rms



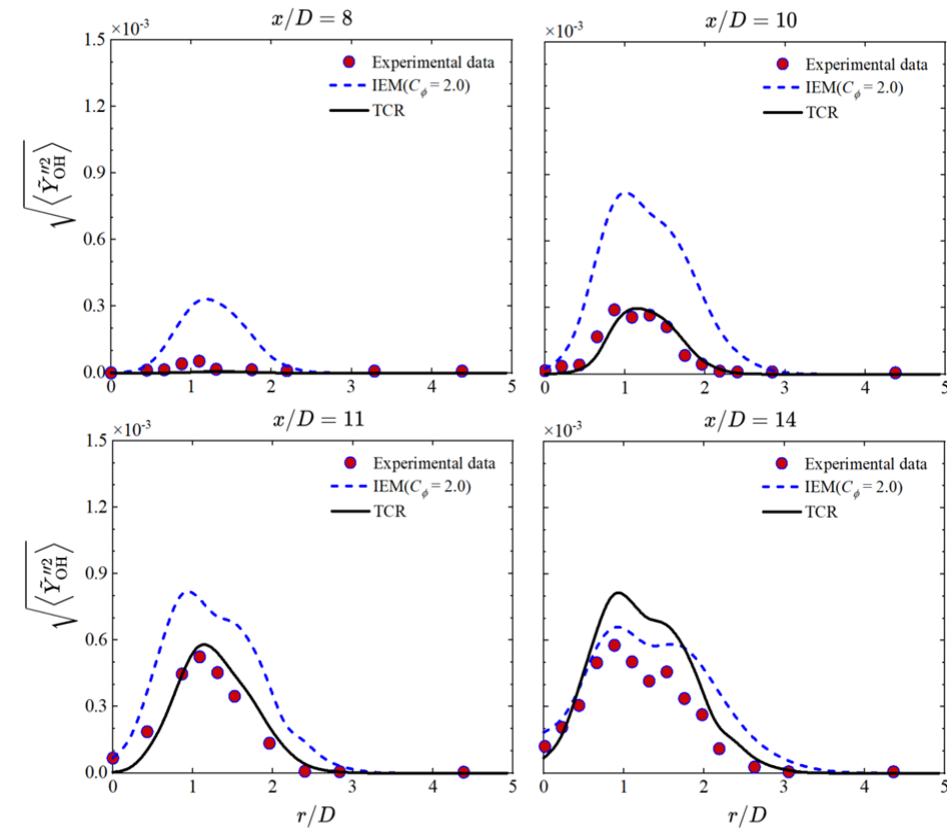
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OH mean



OH rms



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Thank you !